TMFC001 Product Catalog Management

ODA Component Conformance Profile TMFC001 Version – v 1.0.0

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1. Introduction

The conformance requirements outlined in this profile apply to the specified CTK version 1.0.0 of the TMFC001 ODA Component. Any deviations from these requirements may result in non-conformance.

2. Core Functionality

2.1 Compliance Requirements to Exposed API(s)

The below table lists the mandatory Exposed API for TMFC001

API	Version	Reference to CTK
TMF620	4.0	https://www.tmforum.org/oda/open-apis/table

2.1.1 TMF620 version 4.0 API Conformance

The component's API must support all mandatory resources, operations, attributes, and notifications as mentioned in TMF620B (TMF620v4.0 Conformance Profile)

The below sections specify the additional mandatory resources, operations, attributes and notifications to be supported for ODA specific in addition to TMF620 specific API Conformance.

2.1.2 ODA Specific TMF620 API Conformance v4.0

None

2.2 Dependent API Conformance

None

3. Canvas Conformance

3.1 Deployment Conformance

Canvas should be Kubernetes based and it is required to have the component deployment in a Kubernetes based environment. Cluster should be running on a supported Kubernetes version. Ensure that the Kubernetes manifests, deployment configurations, and custom resources are compatible with the targeted Kubernetes API version. Compatibility with 3 previous Kubernetes versions must also be considered for backward compatibility.
only trusted container images from reputable sources must be used.
The component deployment and the Kubernetes cluster must pass the following tests:

1. Step 0: Basic environment connectivity tests
    a. Kubectl configured correctly
        i. The purpose of this test is to check if the kubectl is configured correctly.
        ii. The configuration must be available and the context must be set to the correct cluster
        iii. Kubectl should return pods in <namespace of components> namespace

2. Step1: Deployment component tests
    a. Component can be found in namespace: <namespace of components>
        i. The component must be found in the established namespace for components
    b. Component has deployed successfully (status: Complete)
        i. The component must have deployed successfully and its status must be complete
    c. Test if all exposed api are accessible and return status is 200
        i. All exposed apis defined in the component must provide a valid url
    d. Security api must return at least one partyrole with canvas system role defined in component file
        i. The security api must return at least one partyrole, unless only canvasSystemRole is defined
    e. CTKs for all exposed apis have been executed successfully
        i. This step configures the api ctks. There must be no errors during the process

3.2 Configuration Conformance

Helm chart MUST be used to deploy the ODA Component in a Kubernetes cluster and should contain all necessary resources. Configuration file MUST in YAML.
Namespace MUST exist for Canvas and Components.
Custom resource Definition (CRD) MUST exist for Components and API definitions.
Canvas operator and Canvas component versioning webhook MUST be running.
The deployed component must pass the following configuration checks:

6. Step 0: Component file checks
    a. Component's helm manifest file must exist at the path specified in ctkconfig.json – as retrieved from Kubernetes
    b. File contains valid YAML
        i. Component manifest must be valid YAML
6. Step 1: Component manifest checks
    a. Document of kind 'Component' is found
        i. Component manifest must contain a document of kind: Component
    b. Component api version is within supported versions
        i. Component manifest must contain a supported api version (oda.tmforum.org/v1)
    c. Component has metadata field
        i. Component manifest must contain a metadata field
    d. Component metadata has name and labels
        i. Component metadata must contain name and label fields
    e. Component has spec field
        i. Component manifest must contain a spec field
    f. Spec has coreFunction with exposed and dependent APIs
        i. Component spec must contain a coreFunction field with exposedAPIs and dependentAPIs
    g. Spec has security function
        i. Component spec must contain a security field
    h. Security function has canvas system role or exposed apis
        i. Security function must contain a canvas system role (string) or expose a partyRole API
    i. All resources are labelled with the component name
        i. All resources in the component manifest must be labelled with the component name
    j. Standard component specification exists in the component ctk
        i. TM Forum standard component specification must exist in the resources folder of the component CTK (gets downloaded by the CTK if doest exist)
    k. Component ID from the manifest of component under test matches the standard specification
        i. Component manifest ID must match the ID in the standard component specification
    l. Exposed Apis defined in standard component specification must be specified in component manifest
        i. All mandatory Exposed APIs defined in the standard component specification must be specified in the component manifest

m. Dependent APIs defined in the standard component specification must be specified in the component manifest
    i. All mandatory Dependent APIs in the standard component specification must also be declared in the component manifest
n. All swagger urls must be valid and accessible and version fields
    i. All swagger urls must be valid and accessible

4. Security Conformance

The below table lists the Mandatory Exposed and Dependent API required for Security Conformance of TMFC001

4.1.1 Security Function Static Role Conformance

4.1.1.1 If the security function implements a static role then it must specify a canvasSystemRole

4.1.2 Security Function Dynamic Role Conformance

4.1.2.1 If the security function implements a dynamic role then it must pass Security Exposed API Conformance

If TMF669 is implemented:

API	Version	Reference to CTK
TMF669	4.x	https://www.tmforum.org/oda/open-apis/table

4.1.2.2 **TMF669 API Conformance** v4.x

The component's API must support all mandatory resources, operations, attributes, and notifications as mentioned in TMF669B (TMF669v4.x Conformance Profile)

4.1.2.2.1 ODA Specific TMF669 API Conformance v4.x

None

4.1.2.3 Security Dependent API Conformance

None

5. Management Conformance

Below table lists the Mandatory Exposed and Dependent API required for Management Conformance of TMFC001

5.1.1 Management Exposed API Conformance

None

5.1.2 Management Dependent API Conformance

None

6. Sample CTK Config file – CHANGE_ME.json

6.1CHANGE_ME.json - Parameter Reference Table

Parameter	Description	Editable by User	Required
releaseName	Helm release name of the deployed component under test in the Canvas	Yes	Yes
component_to_run	Component ID of the component under test	Yes	Yes
component_namespace	Canvas namespace of the deployed component	Yes	Yes
standardComponentPath	Override directory location for the TM Forum standard component specification YAML	Yes	Optional
ctk_name_mapping	Override mapping for TMF API IDs to their corresponding CTK folder names if they differ from default	Yes	Optional
runExposedOptional	Flag for running API CTKs for optional exposed apis	Yes	Optional
runDependentOptional	Flag for running API CTKs for optional dependent apis	Yes	Optional
runSecurityOptional	Flag for running API CTKs for optional security apis	Yes	Optional
ctk_download_urls	JSON file reference url with mapping for download paths of required API CTKs	No	Yes
standardComponentDownload{}	Download parameters for standard component specification	No	Yes
ctkconfig.companyName	Name of the company running the conformance certification	Yes	Yes
ctkconfig.productName	Name of the product implementation being tested	Yes	Yes
ctkconfig.productUrl	Public URL for product information or support	Yes	Yes
ctkconfig.componentUrl	Public URL of the TM Forum Component Directory	No	Yes
ctkconfig.headers	Global headers (e.g., Authorization tokens) to be included in API tests requests	Yes	Optional
ctkconfig.payloads	Override payloads for each API CTK being executed	Yes	Optional
ctkconfig.rejectUnauthorized	If true, CTK will enforce TLS certificate validation for HTTPS requests. If false, self-signed or unverified certificates will be accepted (e.g., for local testing).	Yes	Yes
dependentStubs{}	A mapping of dependent TMF APIs to stub release names and optional headers for the	Yes	Yes (For components with mandatory

Parameter	Description	Editable by User	Required
	component under test. Used to identify pre-installed stub components for dependent APIs and configure authentication if required.		dependent APIs)
bddPayloads{}	Dictionary containing base and target payloads for each BDD test scenario for components under test with mandatory dependent APIs	Yes	Yes (For components with mandatory dependent APIs)
retrySettings{}	Configurable retry settings in use by the component CTK when managing deployments	No	Optional

6.2 CHANGE_ME.json – Example reference for component under test


```
{
  "releaseName": "pc-1",
  "component_to_run": "TMFC001",
  "component_namespace": "components",
  "standardComponentPath": "",
  "ctk_name_mapping": {},
  "runExposedOptional": false,
  "runDependentOptional": false,
  "runSecurityOptional": false,
  "ctk_download_urls": "https://raw.githubusercontent.com/tmforum-rand/TMForum-ODA-Component-
Specification/refs/heads/v1beta4/apiIndex.json",
  "standardComponentDownload": {
    "apiBaseUrl": "https://api.github.com",
    "repoOwner": "tmforum-rand",
    "repoName": "TMForum-ODA-Ready-for-publication",
    "gitUrl": "https://raw.githubusercontent.com/tmforum-rand/TMForum-ODA-Ready-for-
publication/refs/heads",
    "gitBranch": "v1beta4",
    "sslVerify": false
  },
  "ctkconfig": {
    "companyName": "TM FORUM",
    "productName": "REFERENCE EXAMPLE PRODUCT CATALOG",
    "productUrl": "www.tmforum.org",
    "componentUrl": "https://www.tmforum.org/oda/directory/components-map",
    "headers": {
      "Accept": "application/json",
      "Content-Type": "application/json"
    },
    "payloads": {
      "TMF669_v4": {
        "PartyRole": {
          "POST": {
            "payload": {
              "name": "qqymucymyd"
            }
```

```
                }
            }
        }
    },
    "rejectUnauthorized": false
},
"dependentStubs": {},
"bddPayloads": {},
"retrySettings": {
    "maxRetries": 30,
    "retryInterval": 10000
}
}
```